

CSA12 - Computer Architecture

Question 1: CPU Simulator

Problem Statement

Design and implement a CPU simulator supporting LOAD, STORE, ADD, SUB, MOV and JMP instructions. Simulate Fetch, Decode, Execute, Memory Access and Write Back stages. Display register and memory contents after each instruction and calculate total execution cycles.

Objective

To understand how a CPU processes instructions by simulating the complete instruction execution cycle.

Theory

A CPU executes instructions in five stages: Fetch, Decode, Execute, Memory Access and Write Back. Registers temporarily store data while memory stores instructions and operands. The Program Counter points to the next instruction.

Algorithm

1. Initialize registers and memory.
2. Read instruction.
3. Fetch instruction.
4. Decode instruction.
5. Execute operation.
6. Access memory if needed.
7. Write result back.
8. Display registers and memory.
9. Count cycles.
10. Repeat until program ends.

Supported Instructions

Instruction	Purpose
LOAD	Load value into register
STORE	Store register value into memory
ADD	Addition
SUB	Subtraction
MOV	Copy register
JMP	Jump to another instruction

Flow (Text)

Start -> Initialize -> Fetch -> Decode -> Execute -> Memory Access -> Write Back -> Display -> Next Instruction -> Stop

Complete C Program

```
#include <stdio.h>
```

```

int main() {
    int R1=0,R2=0,memory[10]={0},cycles=0;
    R1=20; cycles++;
    R2=R1; cycles++;
    R1=R1+10; cycles++;
    R2=R2-5; cycles++;
    memory[0]=R1; cycles++;
    cycles++; // JMP
    printf("R1=%d\nR2=%d\nMemory[0]=%d\nCycles=%d",R1,R2,memory[0],cycles);
    return 0;
}

```

Sample Output

```

R1=30
R2=15
Memory[0]=30
Cycles=6

```

Time Complexity: $O(n)$

Space Complexity: $O(1)$

Applications

- CPU design learning
- Processor simulation
- Computer architecture laboratories
- Educational tools

Viva Questions

1. What is a CPU? – Central Processing Unit that executes instructions.
2. What are the five instruction cycle stages? – Fetch, Decode, Execute, Memory Access, Write Back.
3. What is the purpose of ALU? – Performs arithmetic and logical operations.
4. What is Program Counter? – Stores address of next instruction.
5. Why count execution cycles? – To measure CPU performance.